



PANULIRUS ECHINATUS POPULATION IN THE BRAZILIAN OCEANIC ECOSYSTEM ROCAS ATOLL

BY

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ABSTRACT

In Rocas Atoll (03°51'S 033°48'W) a most abundant population of *Panulirus echinatus* Smith, 1869 was observed during free dives. Numbers of males and females were almost the same, with an overall sex ratio M : F = 1.1. In addition, males are larger than females, with a mean male carapace length (CL) of 92.6 ± 1.8 mm, while the female mean CL was 72.7 ± 1.0 mm and for both sex combined the mean CL was 83.1 ± 1.4 mm. The percentage of females breeding, i.e., with a spermatophore mass or eggs, was 75.1%. More information is needed to better understand the fluctuation in densities over a year cycle, the reproduction period, and differences in CL between males and females, as well as other aspects of the biology of this species. The present work and future information could help to formulate fisheries policies aimed at protecting *P. echinatus* stocks.

Key words. — Spiny lobsters, sex ratio, carapace length, breeding females

RESUMO

No Atol das Rocas (03°51'S 033°48'W) foi observado por mergulhos a população mais abundante de *Panulirus echinatus* Smith, 1869. Havia quase o mesmo número de machos e fêmeas com uma razão sexual total M : F = 1,1. Além disso, os machos são maiores que as fêmeas com média de comprimento de carapaça (CC) de $92,6 \pm 1,8$ mm, as fêmeas de $72,7 \pm 1,0$ mm e para ambos os sexos $83,1 \pm 1,4$ mm. A porcentagem de fêmeas reprodutivas, com presença de massa espermatofórica ou ovos, foi de 75,1%. Mais informações são necessárias para entender melhor a flutuação na densidade ao longo de um ciclo de um ano, período de reprodução, diferenças no CC entre machos e fêmeas e outros aspectos biológicos desta espécie. Este trabalho e informações futuras podem ajudar a formular políticas de pesca para proteger os estoques de *P. echinatus*.

Palavras-chave. — Lagostas, proporção sexual, comprimento de carapaça, fêmeas reprodutivas

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INTRODUCTION

In the most general sense, the vital parameters to characterize a population are growth, survival, age of maturity, fecundity, distribution, and abundance (Begg et al., 1999). These factors have thus been established as instrumental for the recognition and initial delimitation of geographical areas that can be considered representative of stocks (Begg et al., 1999). In addition, the horizontal and vertical distribution patterns exhibited by individuals during the various developmental stages in their life cycle, vary according to biotic and abiotic factors as well as to the specific ecosystems in which they occur (Butler & Herrnkind, 2000; George, 2005a; Mendes et al., 2015). In insular ecosystems, the multiple interactions between physical environmental factors (bathymetry, topography, morphology, platform extension, plate tectonics, and geographic distance) associated with variable patterns of ocean circulations at different scales, as well as species-specific ecological factors, are at the origin of ecological diversity, which influences the structure and dynamics of marine populations (George, 2005a, b; Butler et al., 2011). For species that colonize shallow plateaus around oceanic islands, the intuitive conclusion is that these marine populations are self-sustaining, leading to the inference that there must be mechanisms to favour larval retention (Sinclair, 1988) and consequently self-recruitment (Butler et al., 2011; Cruz et al., 2015). These mechanisms of retention seem to comprise an association between physical processes and active behavioural manifestations of the species (Butler et al., 2011; Iacchei et al., 2016). The physical discontinuity is a dominant factor in the retention of the pelagic phase in the life cycle of the species on oceanic islands, thus allowing the population to persist (Sinclair, 1988). In these marine environments, the structure, abundance, and richness of populations are clearly a function of the existing spatial boundaries (Floeter et al., 2008; Briggs & Bowen, 2012; Kulbicki et al., 2013).

Among the reef invertebrates, spiny lobsters (genus *Panulirus* White, 1847) stand out as key predators of the benthic ecosystem and have one of the greatest dispersion potentials (Cruz et al., 2015). They are ecologically dominant with the same characteristics that make them commercially valuable, i.e., their size and their abundance (Lipcius & Eggleston, 2000). The ecological relevance of these animals stems from their role as a top-chain predator in benthic habitats, and as a prey for sharks, octopuses, and groupers of the pelagic habitat (Tegner & Levin, 1983; Cruz et al., 1986; Edgar, 1990; Eggleston et al., 1990; Lipcius & Eggleston, 2000; Butler & Herrnkind, 2000). On the other hand, these animals also constitute a true highlight for the global economy, since they are the target of one of the largest and most profitable fisheries in the world, due both to their high nutritional value and to their refined use in the human food sector (Gaeta et al.,

2018). Brazil has a prominent position in this respect, occupying the third place in the fishing and commercialization of lobsters on the foreign market, but has also suffered from serious population declines over the last decades due to disorderly and irresponsible fishing (Santana et al., 2016).

There are a few ecological studies available for *Panulirus echinatus* Smith, 1869. In Rocas Atoll, six species of spiny lobsters have been recorded: *P. echinatus*, *Panulirus argus* (Latreille, 1804), *Panulirus laeviscauda* (Latreille, 1817), *Parribacus antarcticus* (Lund, 1793), *Enoplometopus antillensis* Lütken, 1865 and *Palinurellus gundlachi* Von Martens, 1878 (cf. Gaeta et al., 2015). The species *P. echinatus* (known as the “brown lobster”) is the most abundant in Rocas Atoll, and the habitats are diverse, including reef structures and beds of calcareous algae at depths of 0–6 m (Gaeta et al., 2015). Silva et al. (2001) did a first study about sex ratio and total length measurements of this species in Rocas Atoll. Some other studies on *P. echinatus* were conducted in the São Pedro and São Paulo Archipelago (Pinheiro et al., 2003; Pinheiro & Lins-Oliveira, 2006; Góes & Lins-Oliveira, 2009), and yet another study was conducted on biometric relationships of the species in Pernambuco, Brazil (Barreto et al., 2009).

In the present study, we gathered information on *P. echinatus* across the ecosystem around Rocas Atoll, as an initial step towards identification of the population structure and of the habitat use of this lobster. In addition, we also describe some aspects of the biology of *P. echinatus*.

MATERIAL AND METHODS

Site description and habitat

Rocas Atoll (03°51'S 033°48'W) (fig. 1) is the only atoll in the South Atlantic and one of the smallest in the world (Silva et al., 2001). In 1979, this atoll was the first area to be established as a marine protected area in Brazil. Rocas Atoll and other oceanic islands along the coast of Brazil are of great importance for the territorial policy of the Federal Republic, due to their remote location off the continental coast, resulting in an increase of the Brazilian Exclusive Economic Zone. The Marine Reserve includes an area of 360 km², located 266 km off the nearest mainland coast (Pereira et al., 2013). Sediments deposited in the interior of the atoll reefs consist almost entirely of skeletal carbonate, and the development of the sedimentary facies is mediated by the interplay between the sediment source supply, its physical properties, and several hydrographic processes that redistribute and settle the sediments (Maxwell et al., 1964; Milliman, 1974). The predominant sediment inside the atoll is coarse sand, followed by medium sand, and inside the tide pools the sediment ranges from coarse to very coarse, which can be



Fig. 1. Location of Rocas Atoll off the Brazilian coast in the South Atlantic Ocean.

explained by the high hydrodynamism and/or insufficient mud production in this atoll (Pereira et al., 2013).

Surveys of size and sex ratio

The data were obtained inside seven pools (see blue areas in fig. 2) formed during low tide in Rocas Atoll during expeditions in April and October 2014 and April 2016. Lobsters were captured at depths of 0 to 6 m during free dives with minimal interference in the ecosystem and each pool was sampled only once on each expedition. Data collection was done by two divers snorkelling inside each tide pool and searching for lobsters in crevices, under shelters, in holes and among slabs during daytime. All specimens were identified in situ as far as possible and, if necessary, photographed for further identification. Identifications were based on the lobster key of Holthuis (1991) and the main characteristics of colour, spine patterns, antenna size, and other characters were used to verify the morphology of each species.

Lobsters specimens were collected by hand or hand net for measurement of the carapace length (CL) of females and males. No limited time or minimum of animals was stipulated because of difficulties to find and capture those animals

Geomorphological Map

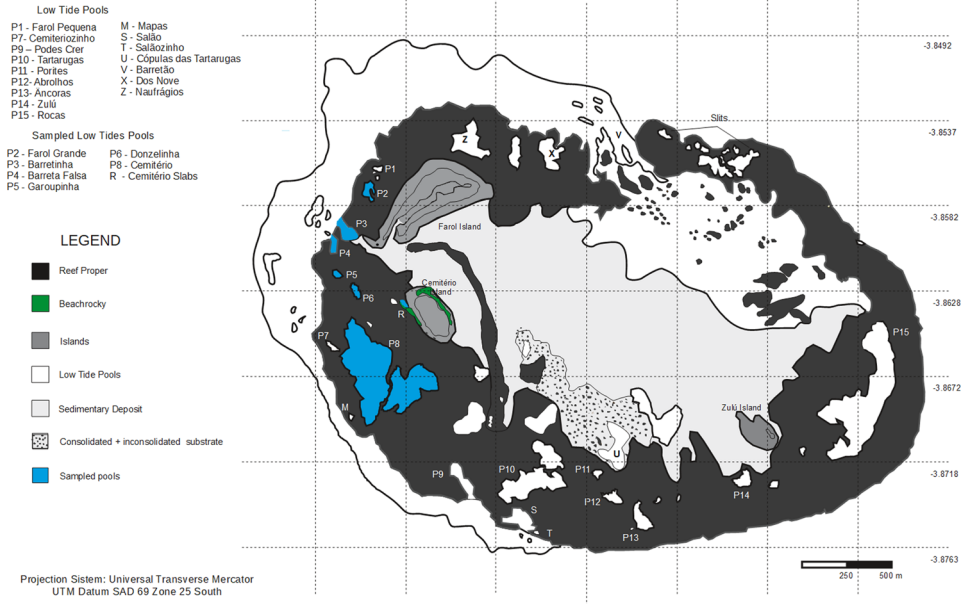


Fig. 2. Tide pools formed in Rocas Atoll at low tide, with the sampled areas indicated.

in each tide pool in this environment. The measurements in situ were made mid-dorsally from between the rostral horns to the posterior edge of the carapace with a Vernier calliper (± 0.1 mm). Each captured animal was sexed and females were also evaluated for the presence of eggs and/or a spermatophore mass and a sample of the reproductive material was taken for later analysis. All animals were returned to the environment after data collection. The sex ratio was determined as the proportion of males per female in the various size classes. The specimens' size-frequency distributions were determined in CL size classes (5 mm intervals) for males and females, starting at 38 mm up to and including 119 mm.

An analysis was made to evaluate the normality of the data through performing the Shapiro-Wilk test, and significant statistical differences between sexes were analysed by Student's "t" test. Histograms and other graphics were subsequently made to show the data distributions and to display the differences.

RESULTS

A total of 444 individuals of *Panulirus echinatus* were measured during the study. Males are significantly larger than females ($p < 0.05$) but the "t" test showed no significant statistical difference between the expeditions for both sexes. Over all sampling occasions there were almost the same numbers of males and females

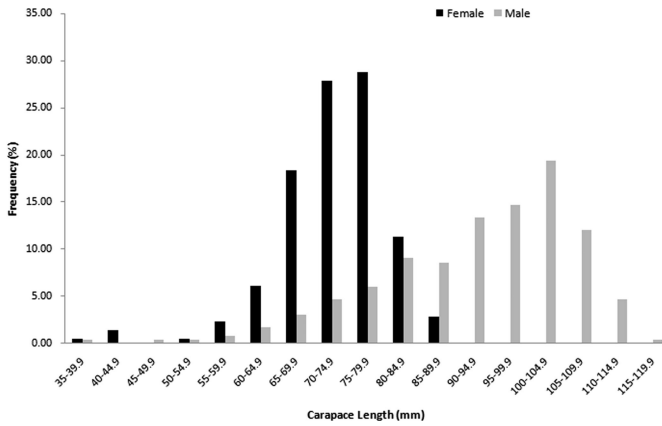


Fig. 3. Distribution of the female and male carapace length classes (mm) of *Panulirus echinatus* Smith, 1869 as observed in Rocas Atoll during the fieldwork in 2014 and 2016.

(overall sex ratio M : F = 1.1). The average male size (carapace length, CL) was 92.6 ± 1.8 mm, while for females that average size was 72.7 ± 1.0 mm, and for both sexes together the value was 83.1 ± 1.4 mm. Females ranged in size from 39.0 to 88.1 mm and males from 38.0 to 118.6 mm. Student's "t" test showed no significant statistical differences between males and females ($p < 0.05$). The more frequent female size class was that of 65-79 mm, while for males the class of 90-105 mm was the most common (fig. 3).

The percentage of breeding females (females with spermatophore mass or/and eggs) was 75.1% of all females. The fact is noteworthy that the majority of females are in the reproductive stage over all carapace length classes, except in the lower size class that represents juveniles (fig. 4). Knowledge of the Size at Onset of Breeding (SOB) is important for determining the percentage of females

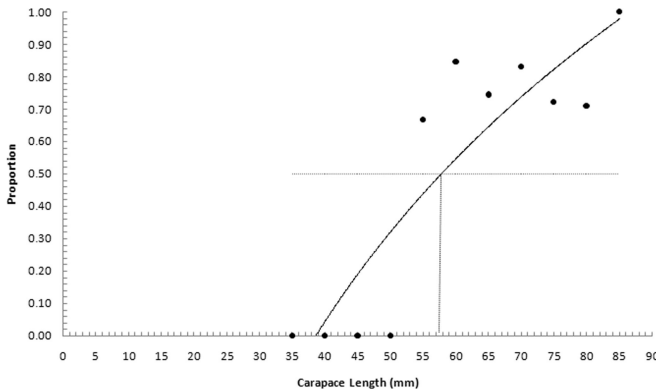


Fig. 4. Proportion of *Panulirus echinatus* Smith, 1869 breeding females per total females in the observed carapace length size classes (mm).

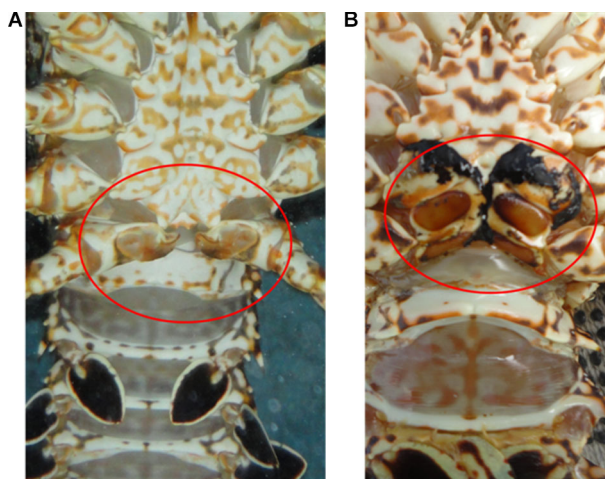


Fig. 5. Reproductive structures of *Panulirus echinatus* Smith, 1869. A, Male with a complex gonopore with an associated pointed and serrated penile process, terminating in a distinct clump of setae; B, female with decalcified windows on the fifth pair of pereopods, separated by calcified ridges.

that contributes eggs to the population pool. SOB is defined as the size where 50% of the females carry eggs. In Rocas Atoll, as a first result, the SOB fell within the size class 55-60 mm CL, with the actually smallest ovigerous female observed being 59 mm in carapace length.

An observation was also made about the animal's reproductive structures, viz., that the male of this species has a complex gonopore with an associated pointed and also serrated penile process, terminating in a distinct clump of setae, while the female has a decalcified window in the base of the fifth pereopod (fig. 5).

DISCUSSION

The *Panulirus echinatus* population we examined at Rocas Atoll shows a small difference in sex ratio, with slightly more males than females. On the other hand, Silva et al. (2001) found a prevalence of females at this same site. Our study and that of Silva et al. (2001), thus deviated from one another in these results, and this may directly be related to the actual collecting procedures, or to the sampling area chosen for analysis, as referred to by Góes & Fransozo (2000). Silva et al. (2001) collected animals during the night at three of the northwestern pools that we observed that have a predominance of females. However, Pinheiro et al. (2003) found in the São Pedro and São Paulo Archipelago the same pattern as recorded by us for *P. echinatus*. Similar patterns of males' predominance were observed in *Panulirus argus* during the months of higher reproductive intensity (Soares, 1994).

The same trend was observed in *P. argus* and *Panulirus laeviscauda* by Coelho et al. (1996). However, Ivo & Neto (1996) also found significant differences in *P. argus*, just as reported by Silva et al. (2001).

The majority of females in our studied area were carrying eggs during the collection of data. Until this moment, no reproductive migration in *P. echinatus* has been reported, so we can not hypothesize that differences found may have been caused by such a migration process, as has been established for *P. argus* (cf. Cruz & Bertelsen, 2009). Thus, it is possible that, by coincidence, we have chosen a spawning zone, and this may obviously influence our results.

Pinheiro et al. (2003) reported that the brown lobster is the smallest palinurid species in Brazil, but we found a mean carapace length (mm) in Rocas Atoll similar to those earlier found for *P. argus* and greater than those reported for *P. laeviscauda* (cf. Lozano-Álvarez & Negrete-Soto, 1991; Vasconcelos et al., 1994; Ivo, 1996; Ivo & Neto, 1996; Pinheiro et al., 2003).

In Rocas Atoll, males exceed females significantly in size above 90 mm CL. Silva et al. (2001) had not found this difference in size between sexes in this atoll, but Pinheiro et al. (2003) found the same relation, indicating a sexual polymorphism in size, based on the lobsters' legs for the São Pedro and São Paulo Archipelago. Kanciruk (1980) assumed that males of palinurids reach considerably greater lengths than females and Vasconcelos & Vasconcelos (1994) and Vasconcelos et al. (1994) observed this also in *P. laeviscauda*. Males would be larger because male growth rate is higher than that of females after a specific size, and/or this phenomenon testifies to a reduced mortality rate for males (Simpson, 1976; Cruz & Phillips, 2000). Moreover, males just spend energy on somatic growth, while females allocate a great amount of energy to reproduction (Hartnoll & Gould, 1988): eggs are more expensive to produce than sperms. Hence, the investment in somatic growth by males guarantees an advantage in intraspecific competition for mating and could explain the larger size observed in males of *P. echinatus*. This same pattern occurs frequently in several groups of decapods that possess in common the incubation of eggs, such as crabs (Brachyura: Grapsidae, and swimming crabs, like Portunidae), hermit crabs (Anomura: Paguroidea), and cleaner and caridean shrimps (e.g., Hippolytidae and Palaemonidae, respectively) (Guerao et al., 1994; Zhang et al., 1997; Pinheiro & Fransozo, 1998; Castiglioni & Santos, 2000; Moura et al., 2000; Mantelatto & Martinelli, 2001).

The increased frequency of large males we found may have implications that come to light in the high percentage of breeding females that we observed in this atoll. In Florida, an increased frequency of large males had important implications for the fecundity of the lobster population (Cox & Hunt, 2005) and it was postulated that the number of females involved in reproduction increased by 30%, and the absolute number increased even more when larger males became

available for mating (Bertelsen & Cox, 2001). The percentage found in our study (75%) for breeding females was similar to the 76 to 89% found by Cox & Hunt (2005) for *P. argus* in different habitats in Florida. A first SOB (Size at Onset of Breeding) estimate of 55-60 mm was reported here for *P. echinatus*. In this respect, it is important to observe that this estimation perhaps does not represent the real SOB, because there was no possibility to sample the breeding cycle of this species throughout an entire year. Nonetheless, this result should be viewed as a first but essential step. Estimates of SOB for *P. argus* have been reported by several studies as ranging from 75 to 85 mm (Cruz & Bertelsen, 2009). Our estimates are lower (55-60 mm) than those reported for *P. argus*. The fact that males are larger than females may influence the SOB size and this may induce an “early” copulation, i.e., a copulation with smaller females.

The absence of ecological and biological data for *P. echinatus* at other sites along the Brazilian coast and at oceanic islands as the Canary and Cape Verde archipelagos, hamper the understanding of fluctuations in density and other differences in the populations of this lobster species. More information will be required to set up effective fisheries policies that can protect *Panulirus echinatus* stocks. Studies in different areas of occurrence of *P. echinatus* are necessary to obtain more complete and more diverse information about this species, and on such a basis make some general fishery rules that would be applicable to this species over its entire distributional range.

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